

KARTHIK NARRAVULA

+91 95029 51059 | nm6773083@gmail.com | linkedin.com/in/karthik-narravula | github.com/Karthik0806 | karthiknarravula.dev

TECHNICAL SKILLS

Languages: Java, C++, Python

Backend: Spring Boot, Spring Security, Spring Data JPA, REST APIs, WebSocket, STOMP, Hibernate, JWT, OAuth2

Databases: PostgreSQL, Redis, MongoDB

Cloud & DevOps: AWS (EC2, RDS, S3), Docker, Nginx, GitHub Actions, CI/CD

Testing: JUnit, Mockito, Integration Testing

Core CS: Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks

PROJECTS

ConnectChat – Real-Time Chat Platform *Java · Spring Boot · WebSocket · STOMP · PostgreSQL · Docker · AWS · OAuth2*

[Live Demo](#) · [GitHub](#)

- Cut message delivery latency to under 100ms by replacing HTTP polling with persistent WebSocket + STOMP connections, enabling private conversations on AWS EC2 + RDS.
- Implemented JWT authentication with access/refresh token rotation and Spring Security filter chains, securing both REST APIs and WebSocket handshakes.
- Integrated Google OAuth2 authentication with Spring Security by implementing secure OAuth callback flows, JWT token generation, frontend redirect handling, and stateless session management across React, Nginx, and AWS deployment.
- Tracked online presence and read receipts in real time by hooking into WebSocket session lifecycle events — users see status updates within milliseconds of connect/disconnect.
- Eliminated N+1 query patterns by designing a normalized PostgreSQL schema with lazy loading and targeted indexing, keeping conversation list queries under 10ms even with large message histories.
- Went from code to live HTTPS endpoint in under an hour by containerizing the backend with Docker and wiring a GitHub Actions pipeline that automatically builds, pushes to Docker Hub, and redeploys on EC2 on every push to main.
- Improved backend reliability by writing unit and integration tests for authentication, messaging workflows, and WebSocket event handling using JUnit and Mockito.
- Documented every endpoint via Swagger/OpenAPI so anyone can test the API at api.karthiknarravula.dev/swagger-ui/index.html.

knurls.me – URL Shortener *Java · Spring Boot · PostgreSQL · Redis · React · JWT · OAuth2 · AWS · Nginx*

[knurls.me](#) · [GitHub](#)

- Built a full URL shortening service from scratch — custom aliases, configurable TTL, and click analytics — and shipped it to [knurls.me](#) without any third-party shortening library.
- Reduced redirect latency for hot links by caching short-code lookups in Redis with a 24-hour TTL, significantly reducing database round-trips for frequently accessed URLs.
- Protected the service against abuse by building a Redis-backed rate limiter using atomic increments, blocking any IP that exceeds 50 requests per minute.
- Captured browser, device type, referrer, and timestamp for every click — giving users a per-link analytics dashboard backed by indexed PostgreSQL queries that return results in under 15ms.
- Added Google OAuth2 alongside JWT + BCrypt so users can sign in with Google or a password — both paths share the same stateless, role-based (USER / ADMIN) security model.
- Hardened the deployment by isolating PostgreSQL and Redis inside a private VPC on AWS, with Nginx handling TLS termination so neither database is ever directly reachable from the internet.
- Validated redirect, authentication, and rate-limiting workflows with automated unit and integration tests, reducing regression risk during deployments.

Patient Service System – Spring Boot · Kafka · gRPC · Eureka · OpenFeign · Spring Cloud Gateway [GitHub](#)

- Explored microservices architecture by decomposing patient management workflows into independently deployable Spring Boot services.
- Implemented service discovery using Eureka and centralized request routing with Spring Cloud Gateway.
- Integrated gRPC and OpenFeign for inter-service communication and experimented with Kafka-based asynchronous event delivery.

EDUCATION

Takshashila University

2023 – 2027

B.Tech in Computer Science & Engineering (OOP, DBMS, OS, CN, COMPILER DESIGN)